

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
ELECTRICAL ENGINEERING DEPARTMENT

EE-Midsem

Monday Apr. 24, 2023
 Time: 15:15-16:00

MS-101 Makerspace
 Spring Semester 2022-23

Marks: 40
 (To be re-scaled)

Roll Number: _____ **Name:** _____

Division: _____ **Batch:** _____ **Signature:** _____

Answer the following in the space provided with the questions.

Q-1 Circuit diagram of the op amp I to V converter used in Expt. 3 is shown below.

The device under test (DUT) is connected between terminals B and C.

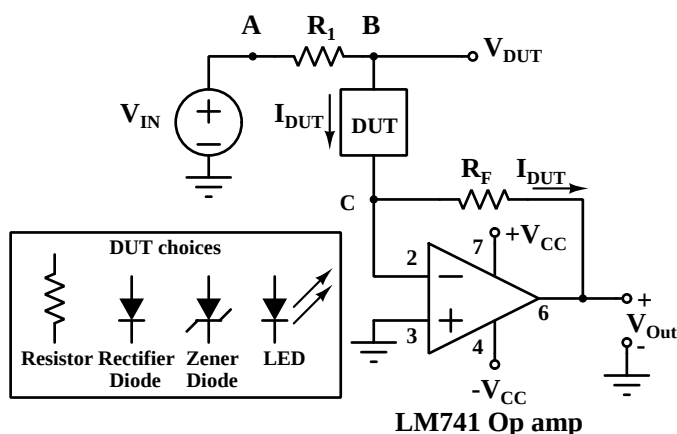
Assume that the op amp is ideal.

$+V_{cc} = +12 \text{ V}, -V_{cc} = -12 \text{ V}; R_1 = 1 \text{ k}\Omega, R_F = 2 \text{ k}\Omega.$

Also assume that the maximum and minimum V_{Out} levels are $+V_{cc}$ and $-V_{cc}$ respectively.

Calculation steps must be shown with your answer.

No marks will be awarded without proper steps.

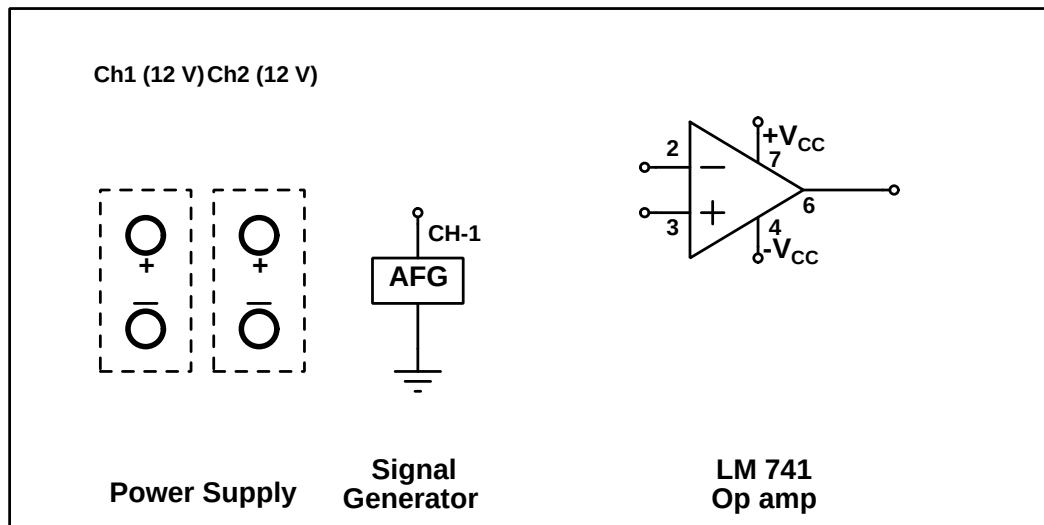


a) A $1 \text{ k}\Omega$ resistor is connected as the DUT.

If $V_{IN} = +4 \text{ V}$, what will be the values of V_{DUT} and V_{Out}?	Steps
Answer:	$V_{DUT} = \underline{\hspace{2cm}} \text{ V},$ $V_{Out} = \underline{\hspace{2cm}} \text{ V}$

– [2]

Q-2 Three blocks with their terminals, *viz.* Power supply, AFG and LM 741 op amp are shown below.



Show the wiring between these three blocks so as to obtain a **non-inverting** amplifier with a voltage gain of **16**. (Add the connections to the diagram above).

Connect also the required resistors out of the following resistor values:

1 k Ω , 10 k Ω , 15 k Ω , 16 k Ω , 17 k Ω , 160 k Ω and 170 k Ω .

Assume that both Ch1 and Ch2 of Power supply are set at 12 V. You must use the Power supply Ch1 as +V_{CC} and Ch2 as -V_{CC}. – [Q2: 5 marks]

Q-3 A digital circuit receives a natural number in the range $0 \leq n \leq 15$ represented by 4 bits ABCD (with A as the most significant bit). We want to design the logic for outputting a ‘Select’ signal when n is any one of 0, 1, 2, 8, 9 or 10.

a)

Fill entries in the truth table shown on the right for ‘Select’ as a function of A, B, C and D. **Be very careful with these entries. Errors in this part will lead to wrong results in the remaining parts!**

Answer:

A	B	C	D	Select
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

– [2]

b)

Express 'select' in canonical form as a sum of products.

Answer:

– [2]

c)

Fill entries in the Karnaugh map shown on the right for 'Select' as a function of A, B, C and D.

Answer:

AB→	00	01	11	10
CD↓				
00				
01				
11				
10				

Show the maximal groupings of '1's in the Karnaugh map above and derive the **minimal** logic expression for 'select' in terms of A,B,C,D and their complements.

Select =

– [4]

– Q3: 2+2+4 = 8 marks

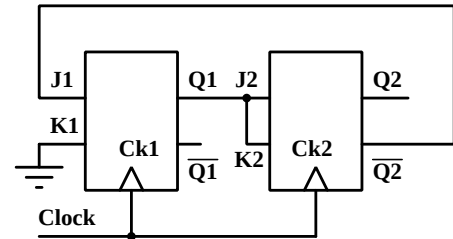
Q-4

We need to generate the function:
 $f = (\overline{P} + \overline{Q}) \cdot (\overline{R} + \overline{S})$ from input signals P, Q, R and S . (Their complements are not provided as inputs).
 IC 7400 provides 4 2-input NAND gates in a single package. Implement the function f using a single 7400 package (*i.e.* Using 4 or fewer 2-input NAND gates).

Logic Diagram:

– Q4: 5 marks

Q-5 Consider the circuit on the right using two positive edge triggered JK flipflops. You are given that at $t=0$, $Q1 = Q2 = 0$.



Four positive clock edges arrive after this initial state. What will be values of Q1 and Q2 after the arrival of each of the clock edges?

	Q1	Q2
At $t=0$	0	0
After Clock1		
After Clock2		
After Clock3		
After Clock4		

– **Q5: 4 marks**

Q-6 a) What is the least number of bits required to represent:

i) 128 (decimal) in unsigned binary format,

Answer:

ii) +128 (decimal) in two's complement signed format,

Answer:

iii) -128 (decimal) in two's complement signed format,

Answer:

iv) +15 (decimal) in two's complement signed format.

Answer:

– [2]

b) i) Represent the unsigned number 1123 (decimal) in base 16 format (Hex code) as a 12 bit wide quantity.

Answer:

ii) Represent the same number (1123 decimal) in base 8 format (octal) as a 12 bit wide quantity.

Answer:

– [2]

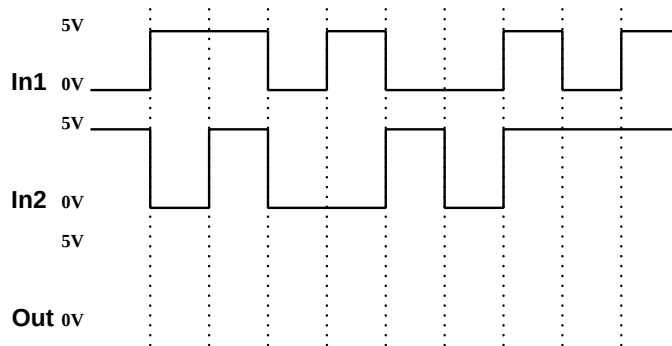
c) What decimal signed number is represented by the 12 bit Hex number D28 if we are using 2's complement convention for signed numbers?

Answer:

– [1]

– **[Q6: 2 + 2 + 1 = 5 marks]**

(Open collector buffers)



– [Q7: 5marks]

Voltage at node B is:	V
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– [Q8: 2 marks]

Paper Ends

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